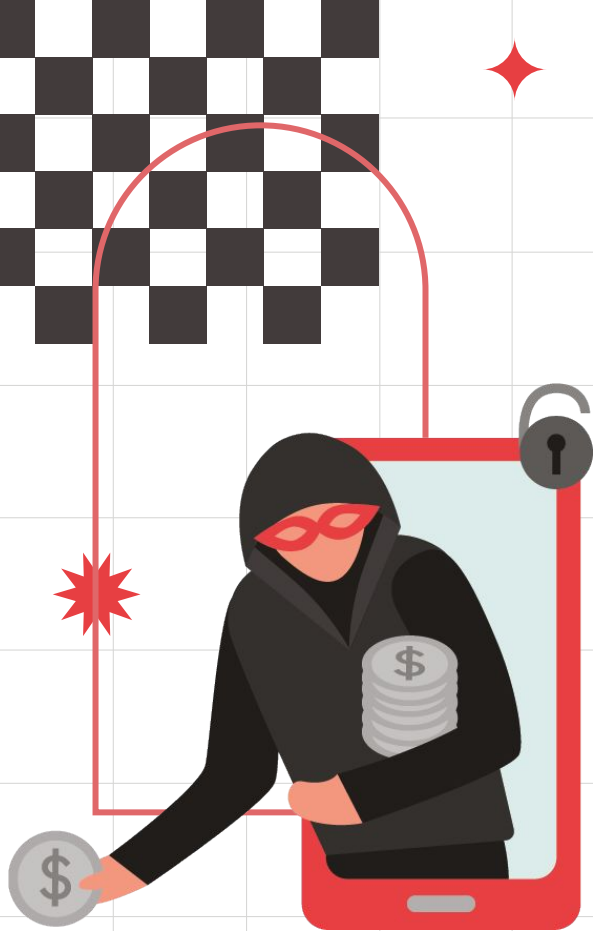




Emotion is All You Need!

Scam Detection with Emotion 

<https://youtu.be/914Wy3iS05k>

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20920347 CHEN Shr En

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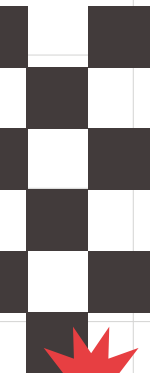
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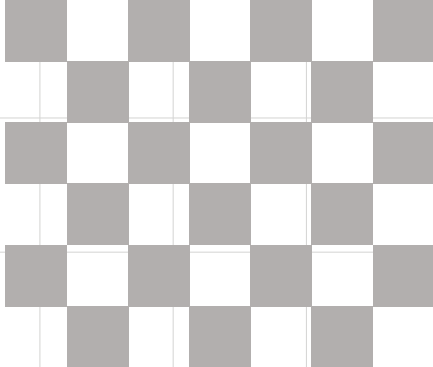
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01

Introduction





Why Emotions for Scam Detection?

- Knowing various of scams from the news, how come people still get tricked?
- Scammers are often good at manipulating victims' emotions, making them susceptible to being tricked.
- Is it possible that there are unique emotion patterns in scam conversations?

WARNING!!

Imminent account closure ⚠

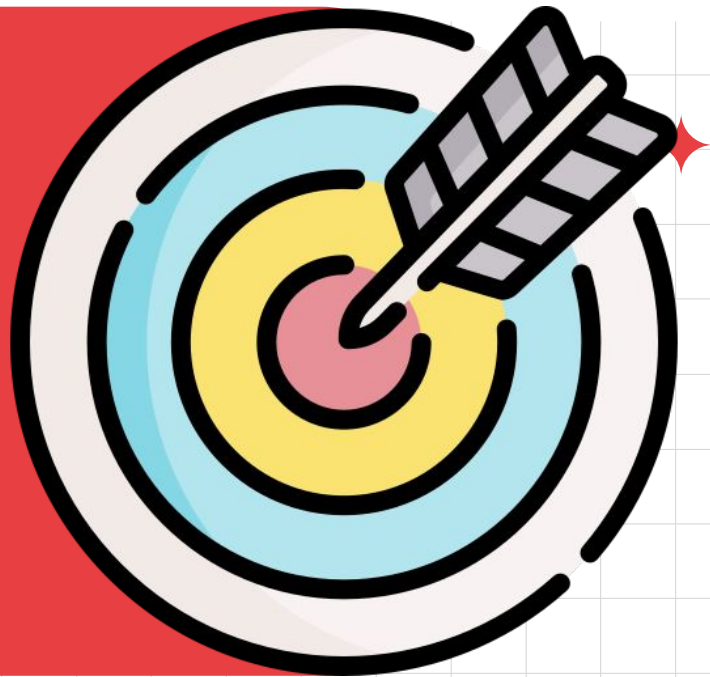
Urgently need money...





Our goal

1. Infer the importance of emotion in scam detection.
2. Improve the scam detection accuracy by integrating emotion patterns as a new feature.





02

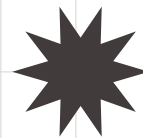
Datasets

Introduction



Datasets Information

Scammer Conversation_[1]



 BothBosu / Scammer-Conversation

conversation
string · lengths



label
int64



Person A: Hello, I'm calling from the bank and need to verify your account details. Person B: I wasn't expecting this call. Can you specify the reason for this verification? Person A: We noticed some unusual activities and need to secure your account. Person B: I understand the concern. Can you provide your employee ID for verification? Person A: Due to security protocols, I cannot share personal details. Person B: I appreciate that, but I need to ensure this call is legitimate. Can you provide an alternate method for me to verify this? Person A: Time is of the essence, and we need to rectify this promptly. Person B: I'd rather take the necessary steps to confirm this independently. Goodbye.

1

Person A: Hi, how was your weekend? Person B: It was great! I visited a beautiful garden and spent time with family. Person A: That sounds lovely. Did...

0

Person A: Hello, this is Sarah from the credit card company. We need to verify your card details. Person B: I wasn't aware of any issues. Can you specify the...

1

Person A: Hey, how's your week going so far? Person B: Hey, not bad, thanks. Just catching up with some reading and planning a weekend trip. How about you?...

0

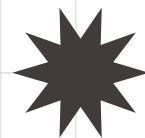
Person A: Good day, I'm calling from a lottery company to inform you that you've won a substantial prize. Person B: Oh, really? Can you provide details on how I...

1

Person A: Hi, how's everything going? Person B: I've been good, thanks. Just finished a home renovation project and it's looking great. How about you? Perso...

0

Datasets Information



Scam dialogue^[2]



dialogue
string · lengths



caller: Hello, is this John? receiver: Yes, it is. Who's calling? caller: My name is Officer Johnson from the Social Security Administration. How are you today? receiver: I'm fine, thank you. What can I do for you? caller: We've been trying to reach you about a very important matter. Your social security number has been compromised and we need to take immediate action to protect your identity. receiver: Oh no, what happened? caller: We've received reports of suspicious activity on your account and we need to verify some information to ensure your benefits aren't interrupted. receiver: Okay, what do you need to know? caller: Can you please confirm your social security number for me? receiver: Wait, I'm not comfortable giving that out over the phone. Is this really the Social Security Administration? caller: Ma'am, I assure you this is a legitimate call. We're trying to help you. If you don't cooperate, we'll have to suspend your benefits. receiver: I don't think so. I'm going to hang up and call the SSA myself to verify this. caller: Ma'am, don't hang up! You need to take care of this now or you'll be sorry!

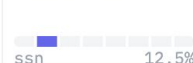
caller: Hello, is this John? receiver: Yeah, that's me. Who's calling? caller: My name is Officer Johnson from the Social...

caller: Hello, is this Mr. Johnson? receiver: Yeah, that's me. Who's this? caller: My name is Officer James from the Social...



BothBosu / scam-dialogue

type
string · classes



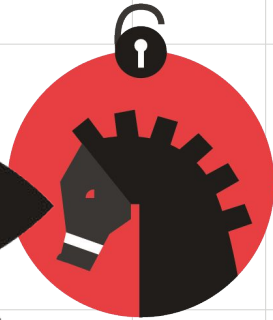
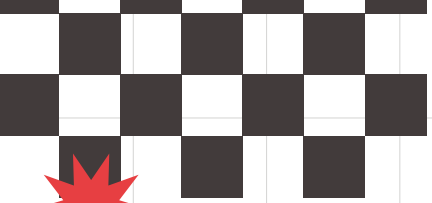
label
int64



ssn 1

ssn 1

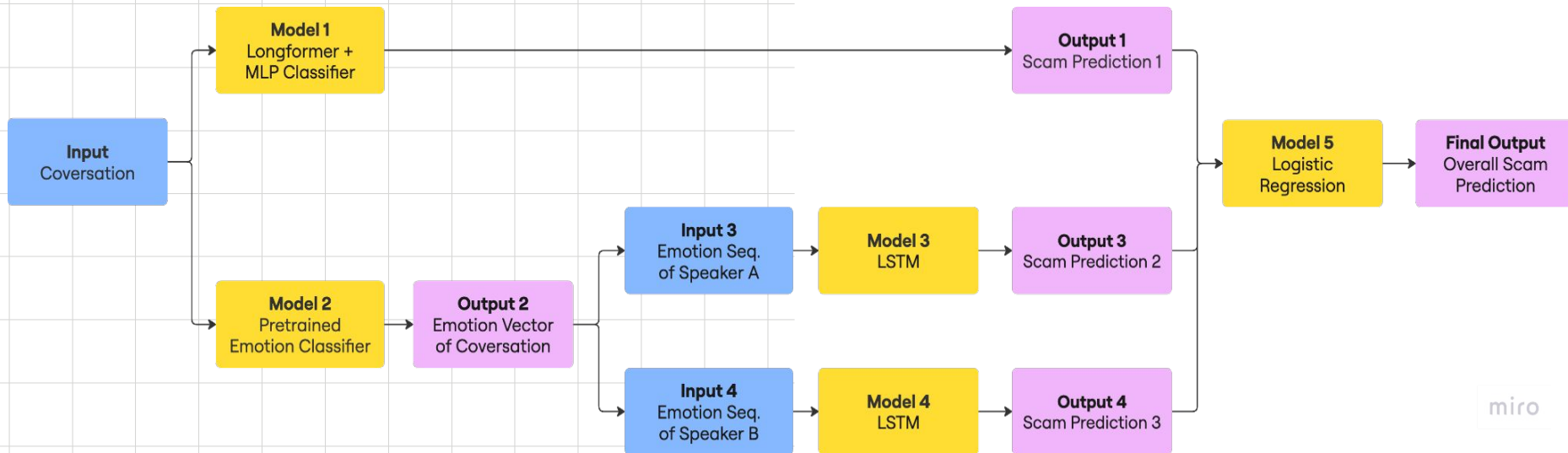
ssn 1



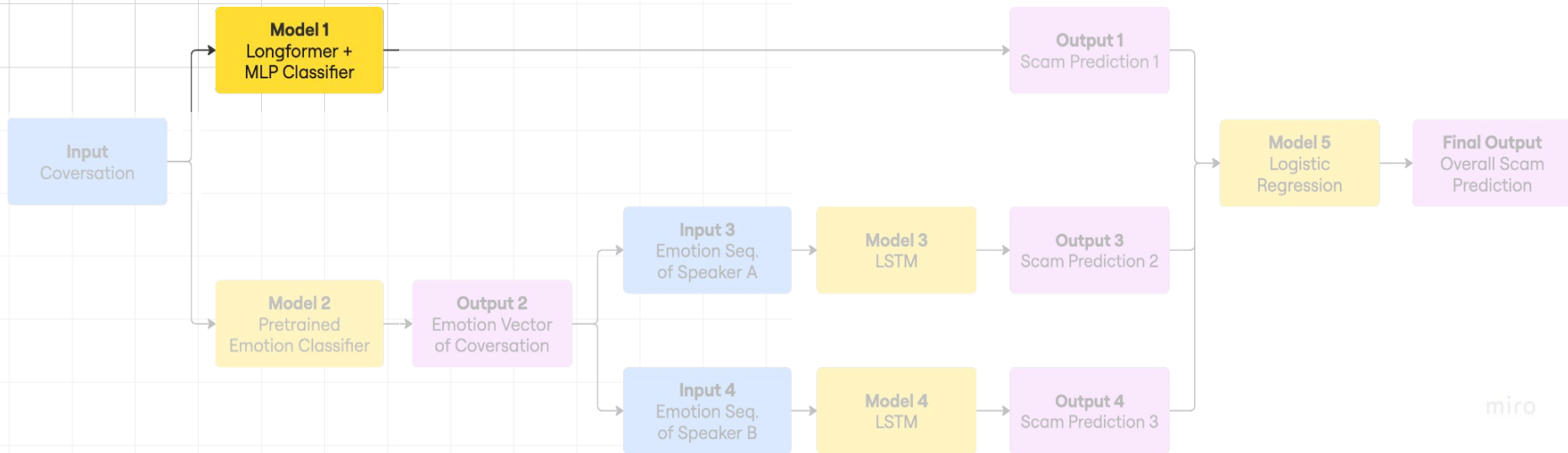
03

Model Architecture

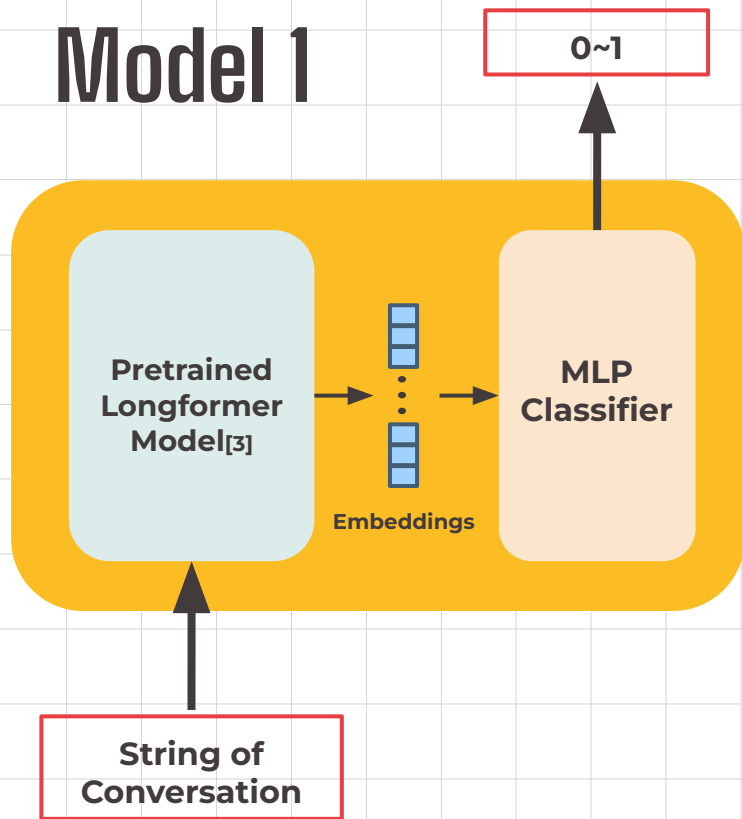
Model Architecture



Model Architecture



Model 1



Function

Predict whether a conversation is a scam conversation.

Training dataset

Preprocessed original dataset (50%).

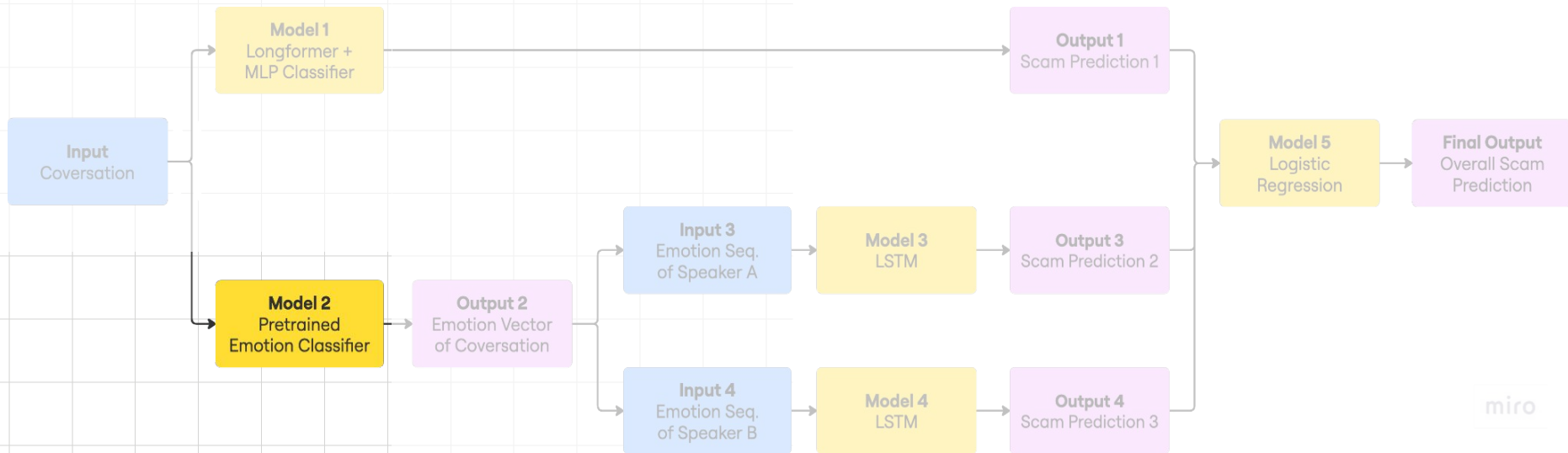
Input

A string of the whole conversation.

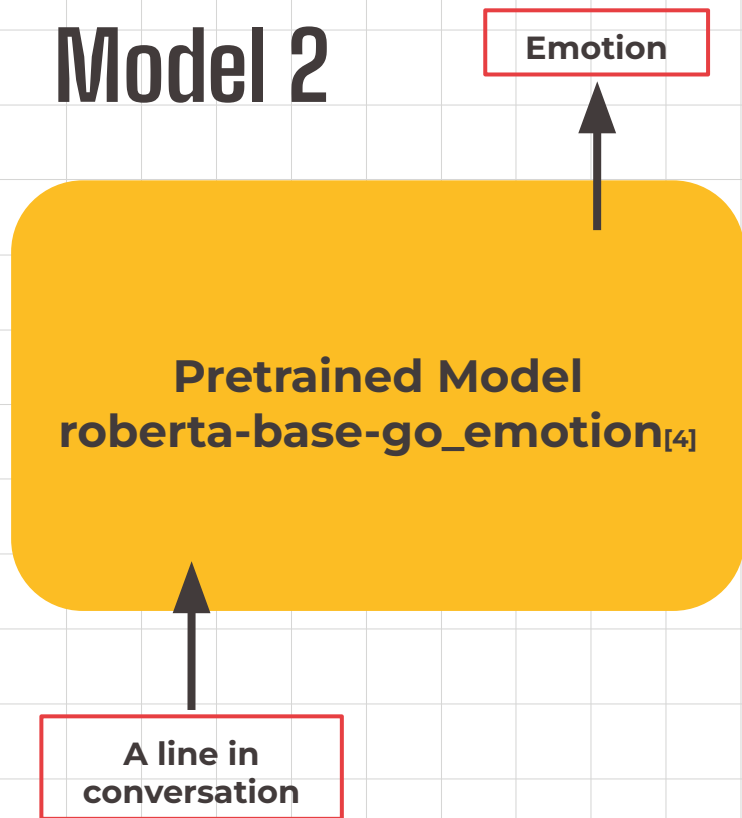
Output

The probability of the input conversation being a scam conversation.

Model Architecture



Model 2



Function

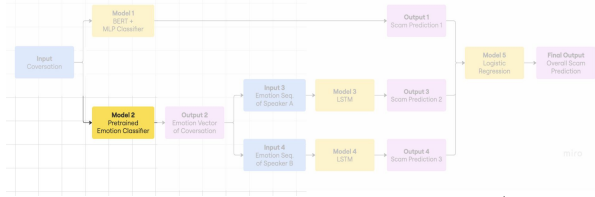
Label the emotion for each line in a conversation.

Input

A string of a line in the conversation.

Output

Possible emotion labels of the line along with the probability.



roberta-base-go_emotions

- A model trained from RoBERTa base model on the go_emotions dataset.
 - RoBERTa base: a transformers model pre-trained on a large corpus of English data
 - go_emotions: A dataset based on Reddit data. It has 28 different emotion labels.

disgust gratitude approval
anger disappointment fear pride
disapproval love excitement
joy sadness relief curiosity
admiration grief confusion annoyance
realization remorse neutral caring
desire nervousness optimism
embarrassment amusement
surprise

⚡ Inference API ⓘ

⚡ Warm ▾

🔗 Text Classification

Examples ▾

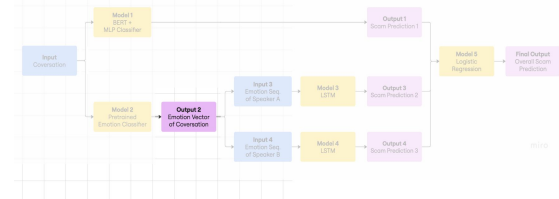
I love machine learning project!

Compute

love	0.935
admiration	0.141
joy	0.021
approval	0.019
neutral	0.009
excitement	0.006
realization	0.005
optimism	0.004
annoyance	0.004
disappointment	0.004
gratitude	0.004
disapproval	0.004
sadness	0.003
amusement	0.003
anger	0.003

Output 2

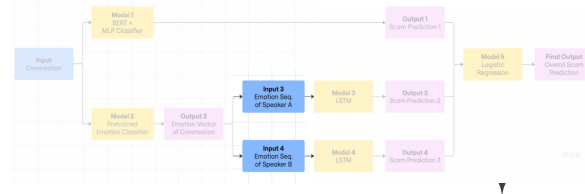
	0	1	2	3	4	5	6	7	8	9	...
0	neutral 0.920	approval 0.025	annoyance 0.010	realization 0.005	desire 0.004	optimism 0.003	excitement 0.003	curiosity 0.003	anger 0.002	caring 0.002	...
1	curiosity 0.662	neutral 0.231	confusion 0.133	surprise 0.093	disappointment 0.013	disapproval 0.011	realization 0.011	annoyance 0.010	optimism 0.005	approval 0.005	...
2	neutral 0.527	realization 0.224	excitement 0.056	approval 0.039	desire 0.024	surprise 0.022	joy 0.013	optimism 0.011	curiosity 0.009	annoyance 0.009	...



- For each line in the conversation, we obtain the probabilities of the line belonging to the 28 emotions in a decreasing order.



Input 3&4



conversation_id	conversation	speaker	emotions	label	one_hot_encoded
0	1597 hello, is this mrs. johnson? yes, it is. who's...	[A, B, A, B, A, B, A, B, A, B, A, B]	[neutral, approval, neutral, curiosity, gratit...	1	[[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
1	104 hi there, how's everything going? i've been gr...	[A, B, A, B, A, B, A, B, A, B, A, B, A]	[curiosity, admiration, approval, curiosity, j...	0	[[0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...
2	377 hi, how have you been lately? i've been great,...	[A, B, A, B, A, B, A, B, A, B, A, B, A]	[curiosity, joy, approval, curiosity, joy, adm...	0	[[0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...
3	1457 hello, this is john from microsoft technical s...	[A, B, A, B, A, B, A, B, A, B, A, B, A]	[neutral, surprise, curiosity, confusion, curi...	1	[[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
4	1427 hello, my name is john and i'm calling from wi...	[A, B, A, B, A, B, A, B, A, B, A, B, A, B, A]	[neutral, surprise, neutral, curiosity, caring...	1	[[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
...	...	...	...	...	...
1135	266 good morning, this is michael from the bank's ...	[A, B, A, B, A, B, A, B, A, B]	[neutral, curiosity, approval, curiosity, neut...	1	[[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
1136	606 hey, how's it going? it's been great, thanks. ...	[A, B, A, B, A, B, A, B, A, B, A, B, A]	[curiosity, admiration, approval, curiosity, j...	0	[[0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...
1137	2156 hi, is this john? no, this is not john. you mu...	[A, B, A, B, A, B, A, B, A, B]	[neutral, disapproval, remorse, approval, admi...	0	[[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...

- Organize the output 2 so the emotion labels of all the lines in a conversation is group together as a list.
- The emotion sequence is then one_hot encoded, becoming a 2D list.



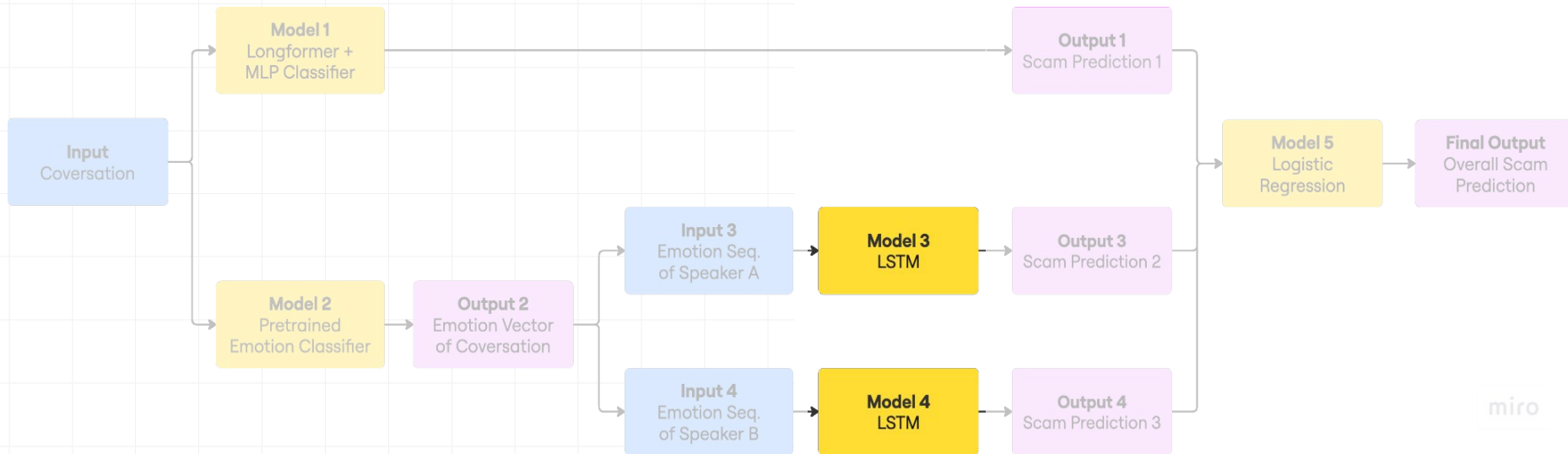
Emotion seq. of conversation 3

Emotion seq. of conversation 3

- # Emotion seq. of conversation 3

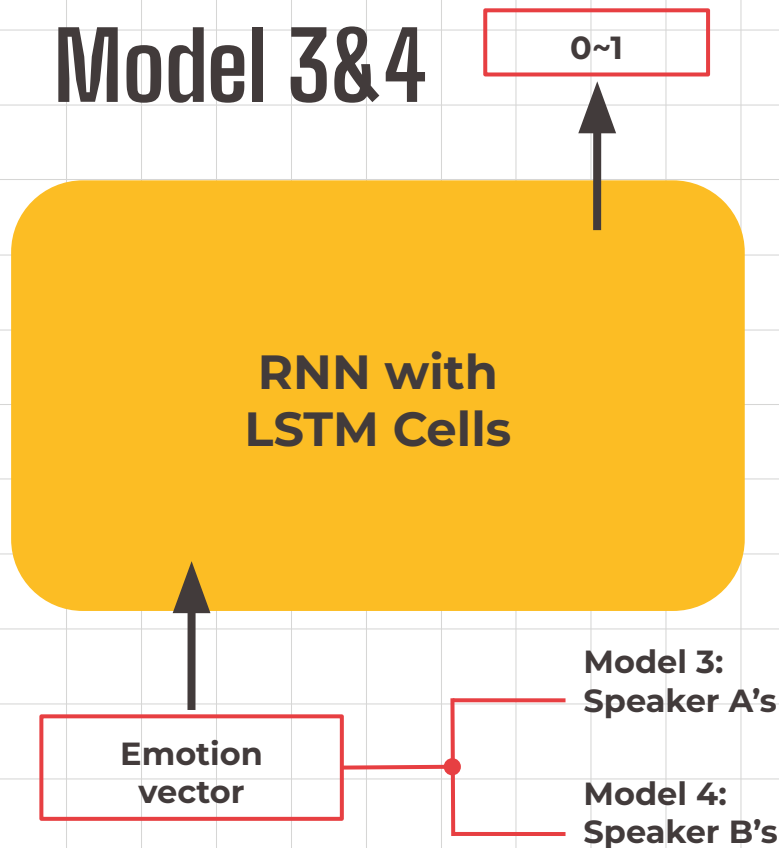
[illegible]

Model Architecture



miro

Model 3&4



Function

Predict whether a conversation is a scam based on emotion sequence extract from the conversation. ✨

Training dataset

Lines in the conversations labeled with emotions by model 2.

Input

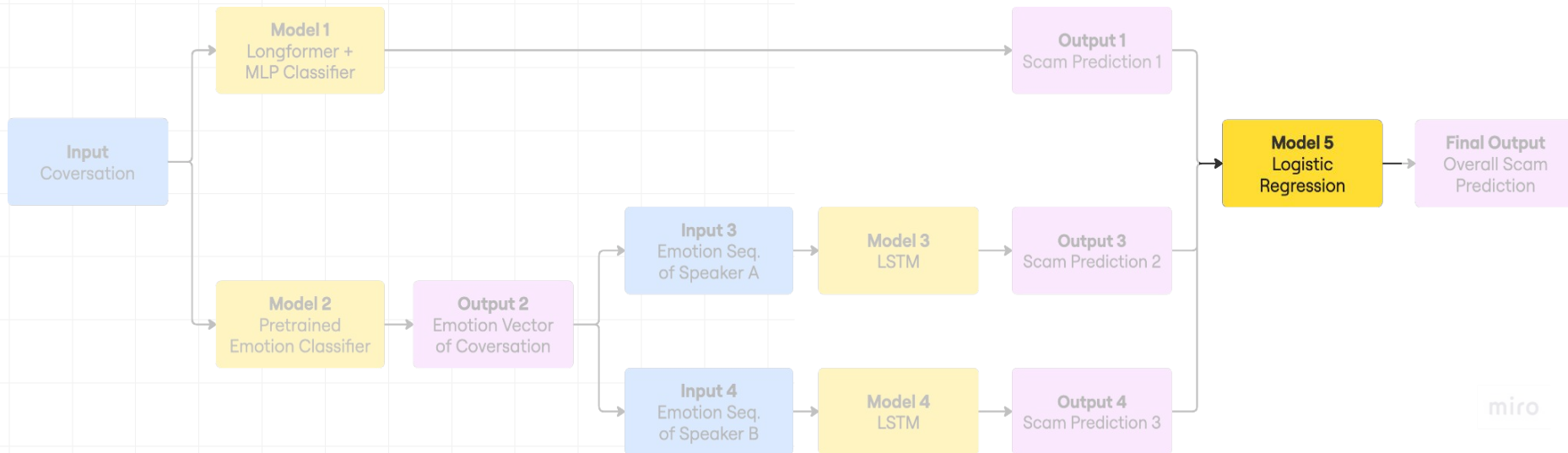
Emotion sequence corresponds to either speaker A's lines or speaker B's lines in a conversation.

Output

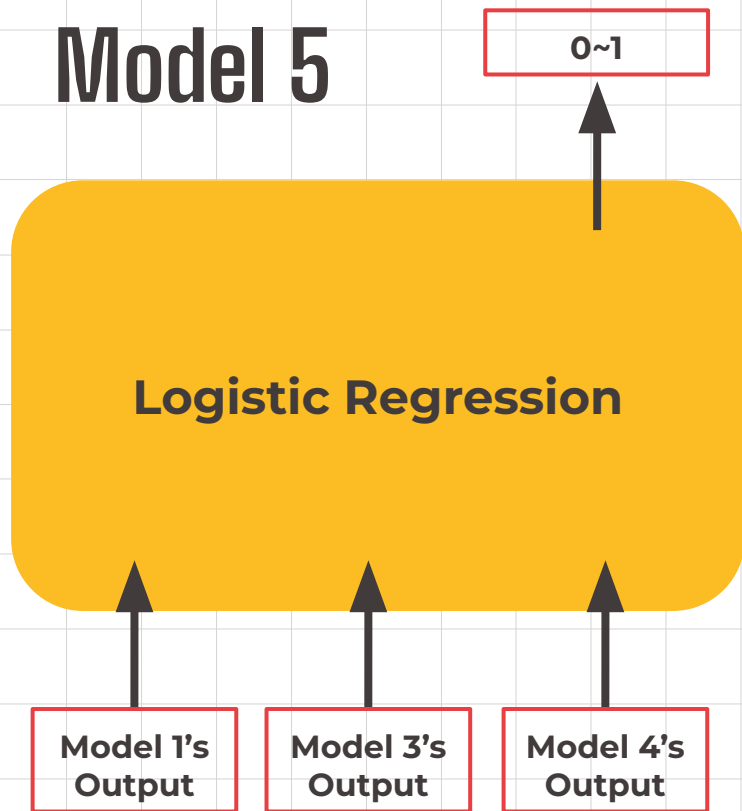
The probability of the input conversation being a scam conversation.



Model Architecture



Model 5



Function

Predict whether a conversation is a scam conversation.

Training dataset

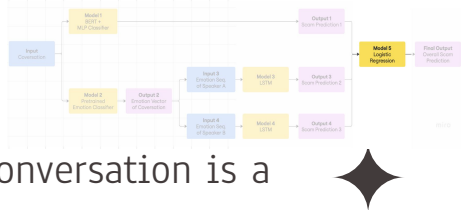
The output of model 1, 3, 4 when inputting data2.

Input

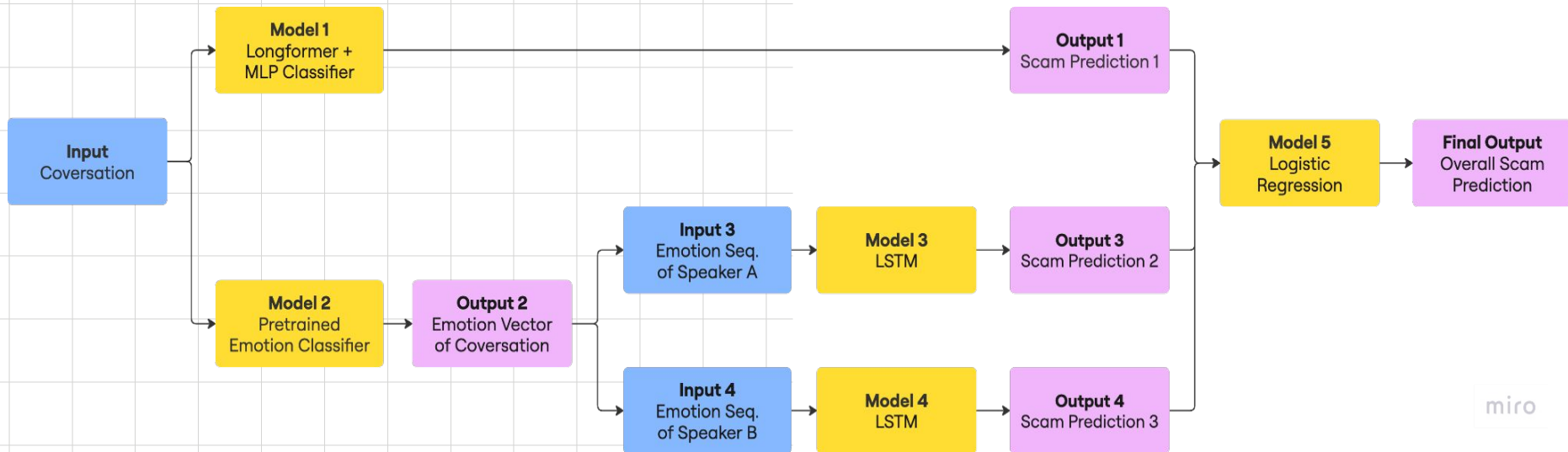
Outputs from model 1, 3, 4 on a conversation.

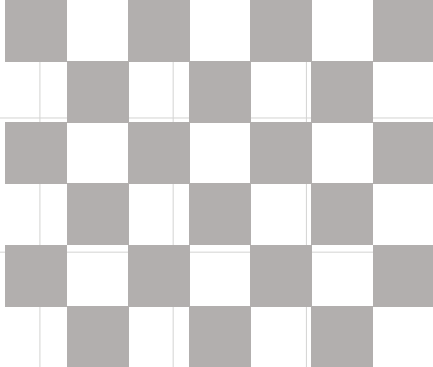
Output

The probability of the conversation being a scam conversation.



Model Architecture





04

Data Preprocessing



Text preprocessing

"Person A: How are you
today? Person B: I'm fine."

"How are you today? I'm fine."

"how are you today? i'm fine."

how are you today? i'm fine.<newline>



Line By Line Splitting

Before

Person A: How are you today? Person B: I'm fine.

After

how are you today?

A

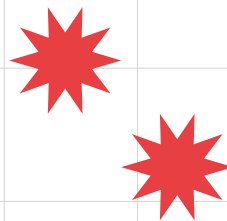
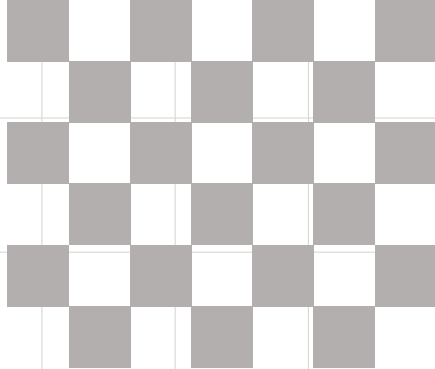
i'm fine.

B

label		conversation	conversation_id	utterance_id	speaker
0	1	hello, is this john?	1000	0	A
1	1	yes, it is. who's calling?	1000	1	B
2	1	my name is officer johnson from the social sec...	1000	2	A
3	1	i'm fine, thank you. what can i do for you?	1000	3	B
4	1	we've been trying to reach you about a very im...	1000	4	A
...	...	...	...	...	...
17465	0	no, i don't know him. but i can try looking up...	2279	7	B
17466	0	okay, thanks so much.	2279	8	A
17467	0	okay, i've got a few listings for john smith i...	2279	9	B
17468	0	that would be great, thank you.	2279	10	A
17469	0	okay, here's one... and here's another one...	2279	11	B

05

Results



Evaluation Method



Evaluate different models

- 1** Model 1
- 2** Model 3
- 3** Model 4
- 4** Model 5



Model 1

Validation Accuracy

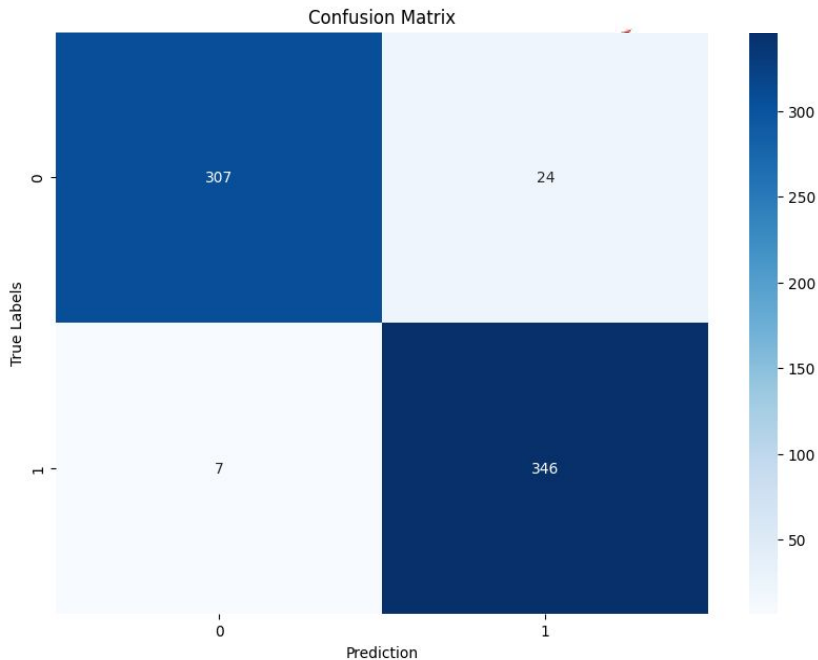
0.9547

F1 Score

0.9546



Confusion matrix



Model 3

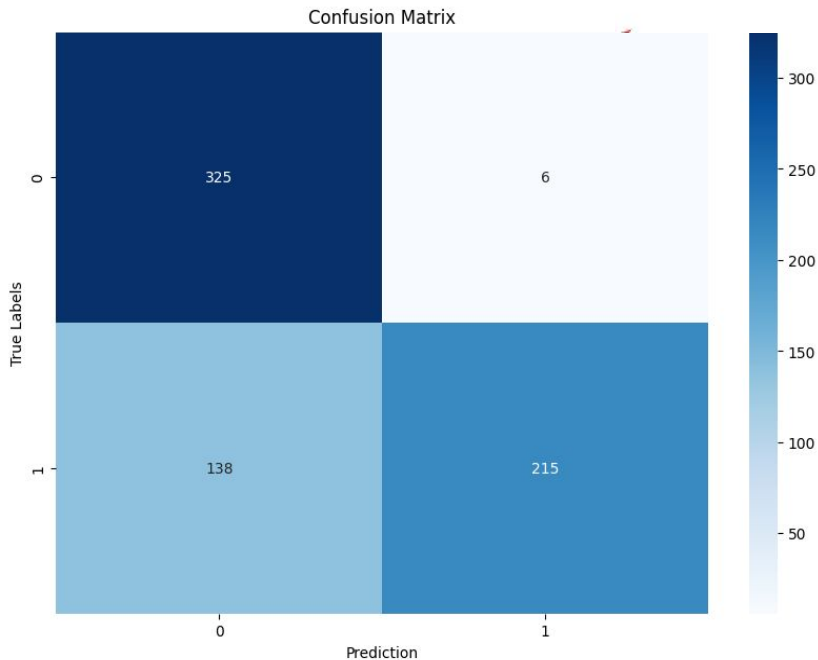
Validation Accuracy

0.7895

F1 Score

0.7828

Confusion matrix



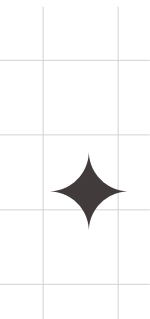
Model 4

Validation Accuracy

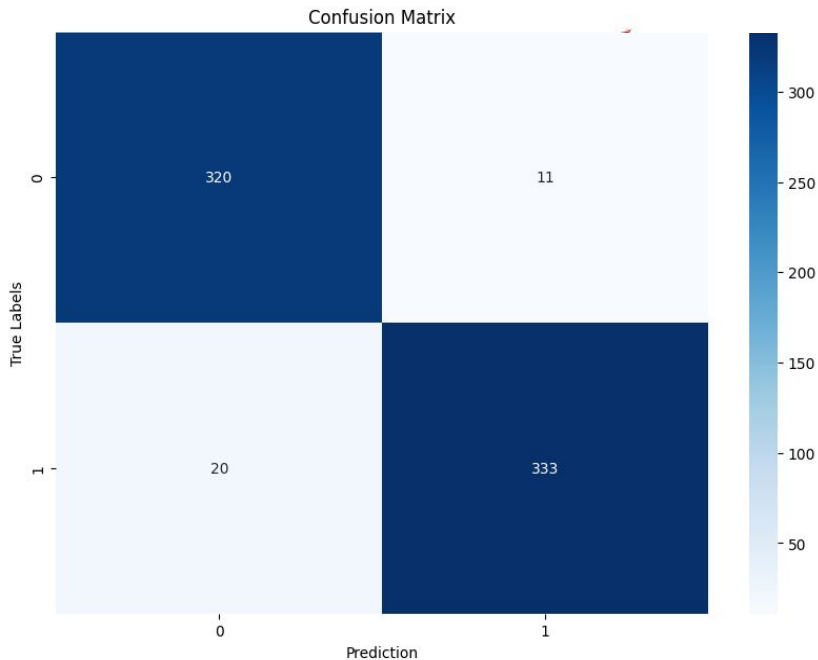
0.9547

F1 Score

0.9547



Confusion matrix



Comparison



Model 1

Scam prediction based
on the conversation text.

**Validation
Accuracy**

95.5%

F1 Score

0.955

Model 3

Scam prediction based
on the speaker A's
emotion sequence.

79.0%

0.782

Model 4

Scam prediction based
on the speaker B's
emotion sequence.

95.5%

0.955

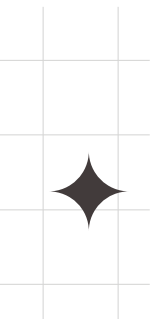
Model 5

Validation Accuracy

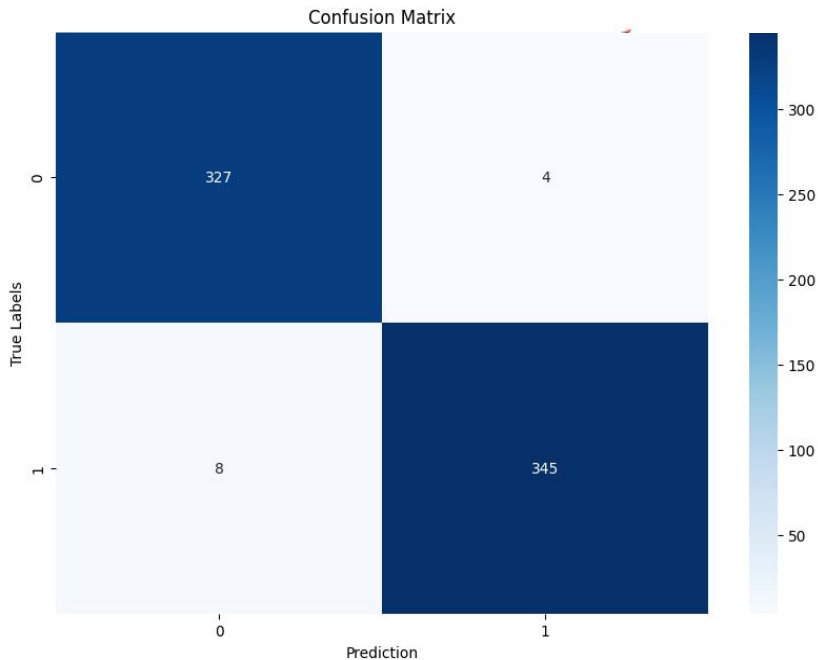
0.9825

F1 Score

0.9825



Confusion matrix



Model 5

Coefficients



	Model 1	Model 3	Model4	Bias
Coefficient	12.62	2.33	6.62	-11.9

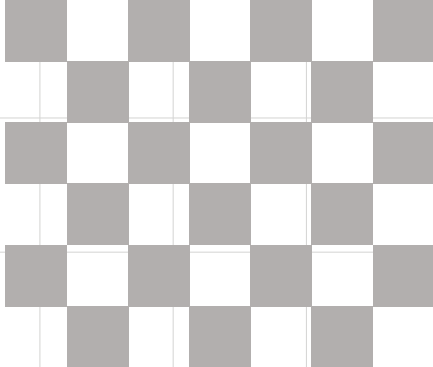
Accuracy Improvement

0.9547



0.9825





06

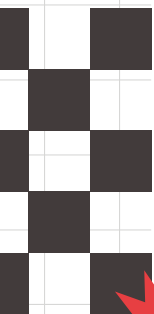
Conclusion



Conclusion



1. Three models were trained for scam detection with the conversation text (model 1) or the emotion patterns of each speaker (model 3&4) respectively.
2. The emotion patterns in the conversation are significant for scam detection, especially the emotion pattern of speaker B.
3. The emotion patterns may be a good reduction of conversation text for scam detection.
4. Combining the three models give the higher accuracy on scam detection compared to the model which solely process on the conversation text.



The background features a light gray grid pattern. There are four star-shaped icons: a large black eight-pointed star in the top-left, a small red four-pointed star in the top-right, a large black eight-pointed star in the middle-right, and a small black four-pointed star in the bottom-left. A large red rectangle is centered horizontally, containing the word "Thanks!" in white. A dark gray horizontal bar is at the bottom, with a large red eight-pointed star in the bottom-left corner.

Thanks!

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)

References

[1][Dataset] BothBosu, "Scam Dialogue Dataset," Hugging Face, 2021. [Online]. Available: <https://huggingface.co/datasets/BothBosu/scam-dialogue>

[2][Dataset] BothBosu, "Scammer Conversation Dataset," Hugging Face, 2021. [Online]. Available: <https://huggingface.co/datasets/BothBosu/Scammer-Conversation>

[3][Model] AllenAI, "longformer-base-4096," Hugging Face, 2021. [Online]. Available: <https://huggingface.co/allenai/longformer-base-4096>

[4][Model] SamLowe, "roberta-base-go_emotions," Hugging Face, 2021. [Online]. Available: https://huggingface.co/SamLowe/roberta-base-go_emotions